

GOVT 6019: Homework 5 — Estimation and Large Samples

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Melissa Sands (Instructor)

Izza Malik (Teaching Assistant)

Part A — Continuous distributions and the normal

Question 1: Arrival times at a polling place

A polling place is open for 13 hours. Let X be the time, in hours after opening, at which a randomly chosen voter arrives. Suppose $X \sim \mathcal{U}(0, 13)$.

- Write down the density $f(x)$ of X . Show that the total area under it is 1.
- What is $\mathbb{P}(X = 5)$? Explain.
- Find the probability that a voter arrives in the last two hours the polls are open. Compute it by hand and with `punif()`.
- Write down the CDF $F(x)$. Find the median arrival time: the x with $F(x) = .5$. Check your answer with `qunif()`.
- Draw 10,000 arrival times with `runif()`. Report the share that fall in the last two hours and compare it with part (c).
- The density of X is $1/13$ on the interval. The density of $\mathcal{U}(0, 0.5)$ is 2 on its interval. Can a probability equal 2? Explain what the height of a density is, if it is not a probability.

Question 2: County turnout

Suppose turnout across counties, in percentage points, is approximately $\mathcal{N}(60, 10^2)$. Let X be the turnout of a randomly chosen county.

- Standardize X at $x = 75$. Use `pnorm()` to find $\mathbb{P}(X > 75)$ twice: once from the standardized value, once by passing `mean` and `sd` to `pnorm()`. Confirm that the answers agree.
- Find $\mathbb{P}(40 \leq X \leq 80)$. Which part of the 68–95–99.7 rule does it correspond to?
- Without R, use the 68–95–99.7 rule to give the share of counties with turnout between 50 and 70, and the share with turnout above 90.
- Use `qnorm()` to find the turnout below which the lowest 10 percent of counties fall. Then find the interval, centered on 60, that contains the middle 95 percent of counties.
- Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Z = (X - \mu)/\sigma$. Use the rules for $\mathbb{E}[a + bX]$, $\mathbb{V}[X + c]$ and $\mathbb{V}[cX]$ from Week 4 to show that $\mathbb{E}[Z] = 0$ and $\mathbb{V}[Z] = 1$.

Part B — Conditional expectation

Question 3: Interest and participation

A survey records each respondent's political interest X (low, medium or high) and the number of political acts Y they took last year, such as contacting an official or attending a meeting. Y is coded 0, 1, or 2 for two or more. In the population, $\mathbb{P}(X = \text{low}) = .40$, $\mathbb{P}(X = \text{medium}) = .45$ and $\mathbb{P}(X = \text{high}) = .15$. The conditional distribution of Y given X is below. (The probabilities are stipulated for the exercise.)

	$Y = 0$	$Y = 1$	$Y = 2$
low	.70	.20	.10
medium	.40	.40	.20
high	.10	.40	.50

- Find $\mathbb{E}[Y \mid X = x]$ for each level of interest.
- $\mathbb{E}[Y \mid X]$ is a random variable. List the values it takes and the probability of each. A respondent has not yet been drawn. What is the probability that $\mathbb{E}[Y \mid X] = 1.4$?
- Use the law of iterated expectations to find $\mathbb{E}[Y]$.
- Find the marginal distribution of Y , $\mathbb{P}(Y = y)$ for $y = 0, 1, 2$, from the table and the probabilities of X . Compute $\mathbb{E}[Y]$ from it and confirm your answer to part (c).

Part C — The variance of a sample proportion

Question 4: The variance of a sample proportion

Let X be a Bernoulli variable with $\mathbb{P}(X = 1) = p$. From Week 4, $\mathbb{V}[X] = p(1 - p)$. A poll records n independent draws of X and reports the sample proportion $\bar{X}$.

- Plot $\mathbb{V}[X] = p(1 - p)$ for p between 0 and 1. At which p is it largest? What is its largest value?
- Show that $p(1 - p) = .25 - (p - .5)^2$. Use this identity to show that $p(1 - p) \leq .25$ for every p .
- Write $\text{SD}[\bar{X}]$ as a function of p and n . Give an upper bound on $\text{SD}[\bar{X}]$ that holds at every p .
- A pollster fixes $n = 1,000$ before knowing p . Compute the bound from part (c). Compute the true $\text{SD}[\bar{X}]$ at $p = .38$ and at $p = .05$. By how much does the bound exceed each, as a difference and as a ratio?
- Simulate 5,000 polls of $n = 1,000$ at $p = .5$, and 5,000 more at $p = .05$. For each p , report the standard deviation of the 5,000 sample proportions. Compare it with your answers to parts (c) and (d).
- What n brings the bound from part (c) down to .01?

Part D — The law of large numbers and the central limit theorem

Question 5: Rolling dice in R

In class, each pair rolled a die four times, recorded the sum of the four rolls, repeated this 40 times, and marked each sum on a dot plot. The axis also showed the mean of the four rolls: the sum divided by 4. Let $\bar{X}$ be the mean of four independent rolls of a fair die. From lecture, $\mu = 3.5$, $\sigma = \sqrt{35/12} \approx 1.708$, and $\text{SD}[\bar{X}] = \sigma/\sqrt{4} \approx 0.854$.

- Write a function `trial()` that simulates one trial and returns the mean of the four rolls. Use `sample()`.
- Run 40 trials. Plot the 40 means with one bar for each possible value of $\bar{x}$. Count how many of the 40 means fall between 2.75 and 4.25 inclusive.
- The values of $\bar{x}$ from 2.75 to 4.25 are exactly those within 0.854 of μ . Enumerate all $6^4 = 1,296$ equally likely outcomes of four rolls with `expand_grid()`, and compute the exact probability that $\bar{X}$ falls between 2.75 and 4.25. Compare it with the probability a normal distribution places within one standard deviation of its mean.
- Let K be the number of the 40 trials that fall between 2.75 and 4.25. Name the distribution of K and give its parameters. Find $\mathbb{E}[K]$ and $\text{SD}[K]$.
- Repeat the 40-trial experiment of part (b) 1,000 times, recording K each time. Plot the distribution of the 1,000 values of K . Report their mean and standard deviation and compare them with part (d). What share of the 1,000 values are 24 or fewer?
- Run 10,000 trials instead of 40. Report the share of the 10,000 means that fall between 2.75 and 4.25. Name the result that describes how this share behaves as the number of trials grows.

Question 6: Small donations

Individual contributions to a campaign, in dollars, follow a lognormal distribution: $X = e^W$ with $W \sim \mathcal{N}(4, 1)$. In R the draw is `rlnorm(n, meanlog = 4, sdlog = 1)`. The population mean is $\mu = e^{4.5} \approx 90.02$ and the population standard deviation is $\sigma \approx 118.00$.

- Draw 100,000 contributions. Plot their histogram and report their mean and median. Describe the shape of the distribution. Explain why the mean exceeds the median.
- For each of $n = 1, 10, 100, 1000$, draw 5,000 samples of size n and record the mean of each sample. Plot the four sets of sample means as four histograms, one per n . On each, overlay the density of $\mathcal{N}(\mu, \sigma^2/n)$.
- For each n , report the mean of the 5,000 sample means, their standard deviation, and $\sigma/\sqrt{n}$. Compare the mean of the sample means with μ , and their standard deviation with $\sigma/\sqrt{n}$. Do they agree at every n ? Which results from lecture guarantee the agreement?
- For each n , report the share of the 5,000 sample means that fall below μ . What would this share be if the sampling distribution were normal? Explain why the share exceeds that value at small n .
- Standardize each sample mean: $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$. For each n , report the share of the Z values with $|Z| \leq 1$, and the 2.5th and 97.5th percentiles of Z . Compare them with `pnorm(1) - pnorm(-1)` and `qnorm(c(.025, .975))`.

- f. In class, the mean of four dice already had a roughly normal distribution. Here the mean of ten contributions does not. Name the feature of the two populations that accounts for the difference.
- g. **Beyond the lecture.** Repeat the simulation of part (b), without the normal overlay, with draws from `rcauchy(n)`, the standard Cauchy distribution. A few Cauchy means are enormous, so plot only the means between -10 and 10 (`filter(abs(xbar) < 10)`), and report the quartiles of the sample means at each n . Describe how the distribution of the sample mean changes with n . The Cauchy distribution has no finite mean and no finite variance. State which assumption of the central limit theorem fails.

Question 7: India's missing girls

Across human populations, the probability that a birth is a girl is stable at about $p = .487$. In India in 2009 there were roughly 15 million births. About 47 percent of them were girls (Gailmard, pp. 202–203).

Model each birth as an independent Bernoulli draw: $X_i = 1$ if birth i is a girl, with $\mathbb{P}(X_i = 1) = .487$. Let $\bar{X}$ be the share of girls among $n = 15,000,000$ births.

- a. Find $\mathbb{E}[\bar{X}]$ and $\text{SD}[\bar{X}]$. Compare $\text{SD}[\bar{X}]$ with the bound from Question 4(c).
- b. Use the central limit theorem to give the approximate distribution of $\bar{X}$. Standardize the observed $\bar{x} = .47$. How many standard deviations does it lie from $\mathbb{E}[\bar{X}]$?
- c. Compute $\mathbb{P}(\bar{X} \leq .47)$ with `pnorm()`. R returns 0. Is the probability exactly zero? Explain what R's answer means. Hint: the smallest positive number R can store is about 10^{-308} .
- d. Simulate 10,000 values of $\bar{X}$ with `rbinom(10000, size = 15e6, prob = .487) / 15e6`. Report the smallest and largest of the 10,000 values and compare them with .47.
- e. The same 47 percent can come from samples of different sizes. For $n = 1,000, 4,000$ and $10,000$ births, compute $\mathbb{P}(\bar{X} \leq .47)$ under the model. What is the smallest n at which .47 lies at least 2 standard deviations below .487?
- f. The model assumes the births are independent. Identical twins are always the same sex. Use the formula for the variance of a sum from lecture to say whether twins make $\mathbb{V}[\bar{X}]$ larger or smaller than $p(1 - p)/n$. By what factor would $\text{SD}[\bar{X}]$ have to grow for .47 to lie 2 standard deviations below .487? Can twins account for the gap?
- g. How many girls are “missing”? Compute two counts:
 - (i) the expected number of girls among 15 million births, minus the observed number;
 - (ii) the number of girls expected alongside the observed number of boys, minus the observed number of girls. The two answers differ by about a factor of two. Explain which assumption about the missing births each count makes.
- h. The calculation shows that 2009's births did not come from a process with $p = .487$. It does not say why. Name two processes that could produce a recorded share of .47.

Part E — Estimators and their properties

Question 8: Five estimators of a share

A pollster wants θ , the share of voters who support a ballot measure. Suppose $\theta = .50$. The poll draws n voters independently, with $X_i = 1$ if voter i supports the measure and $X_i = 0$ otherwise. Consider five estimators:

- $\hat{\theta}_A = \bar{X}_n$, the share in the whole sample;
- $\hat{\theta}_B$, the share among the first 4 respondents;
- $\hat{\theta}_C$, the share among the first 10 respondents;
- $\hat{\theta}_D = \bar{X}_n + 1/n$;
- $\hat{\theta}_E = \bar{X}_n + .05$.

- Find the expected value and the bias of each estimator. Which are unbiased?
- Find the variance of each estimator, as a function of θ and n .
- Rank the unbiased estimators by efficiency, for $n \geq 10$.
- At $n = 100$, compute the MSE of each estimator. Name a biased estimator with a smaller MSE than an unbiased one.
- One poll of $n = 100$ gives $\hat{\theta}_A = .54$. What is the error of this estimate? Does it show that $\hat{\theta}_A$ is biased? Why can a pollster not compute the error of a real poll?
- For each of $n = 10, 100, 1,000, 10,000$, simulate 5,000 polls. For each estimator and each n , report the share of the 5,000 estimates that lie within .02 of θ , that is, with $|\hat{\theta} - \theta| \leq .02$. Which shares go to 1 as n grows? Which estimators are consistent?

Two hints. In R, `0.52 - 0.5 <= 0.02` is FALSE, because decimals are stored with tiny rounding errors. Test `round(abs(est - theta), 10) <= 0.02` instead. To keep the simulation fast, draw the first 10 respondents of each poll one at a time, and draw the number of supporters among the other $n - 10$ with `rbinom(5000, n - 10, theta)`.

Question 9: Dividing by n or by $n - 1$

Week 2 defined the sample variance with the divisor $n - 1$. This question shows why, and what the other divisor would buy. Let $X_1, \dots, X_n$ be iid with mean μ and variance σ^2 . Let $\text{SSD} = \sum_i (X_i - \bar{X})^2$ be the sum of squared deviations from the sample mean.

- Show that $\text{SSD} = \sum_i X_i^2 - n\bar{X}^2$.
- Use $\mathbb{E}[W^2] = \mathbb{V}[W] + (\mathbb{E}[W])^2$ to find $\mathbb{E}[X_i^2]$ and $\mathbb{E}[\bar{X}^2]$ in terms of μ , σ^2 and n .
- Show that $\mathbb{E}[\text{SSD}] = (n - 1)\sigma^2$.
- Two estimators of σ^2 are $s^2 = \text{SSD}/(n - 1)$ and $\hat{\sigma}^2 = \text{SSD}/n$. Find the expected value of each. Which is unbiased? What is the bias of the other?
- Simulate 100,000 samples of $n = 10$ from $\mathcal{N}(0, 1)$, so $\sigma^2 = 1$. For each sample compute s^2 and $\hat{\sigma}^2$. For each estimator, report its bias, its variance and its MSE. Which has the smaller MSE?
- Give one reason to prefer s^2 and one reason to prefer $\hat{\sigma}^2$. What happens to the difference between them as n grows?